Dataset Validation

Surgical Scene Semantic Segmentation

Dataset: dsad

Classes: 8 | Videos: 23 | Frames: 1,430

Class presence is analysed at three levels - frames, pixels and videos - on the complete dataset (no sampling), globally and for every split. Video coverage highlights class imbalance that frame and pixel counts alone can hide.

1. Executive Summary

Dataset: dsad

Dataset

- 8 classes, 23 videos, 1,430 frames.
- train: 863 frames, 15 videos
- validation: 202 frames, 3 videos
- test: 365 frames, 5 videos
- Ignored (out-of-range) pixels: 0

Class imbalance warnings (global, thresholds: <= 3 videos, >= 50% pixels in one video)

- spleen: in 5/23 videos (22%), 75% of its pixels in 12

Classes missing from a split

- validation: spleen
- test: spleen

Dominant / smallest pixel share (global)

- Dominant: background (80.94%)
- Smallest: pancreas (0.2659%)

2. Dataset Overview

Dataset: dsad

Split	Frames	Videos
train	863	15
validation	202	3
test	365	5
global	1,430	23

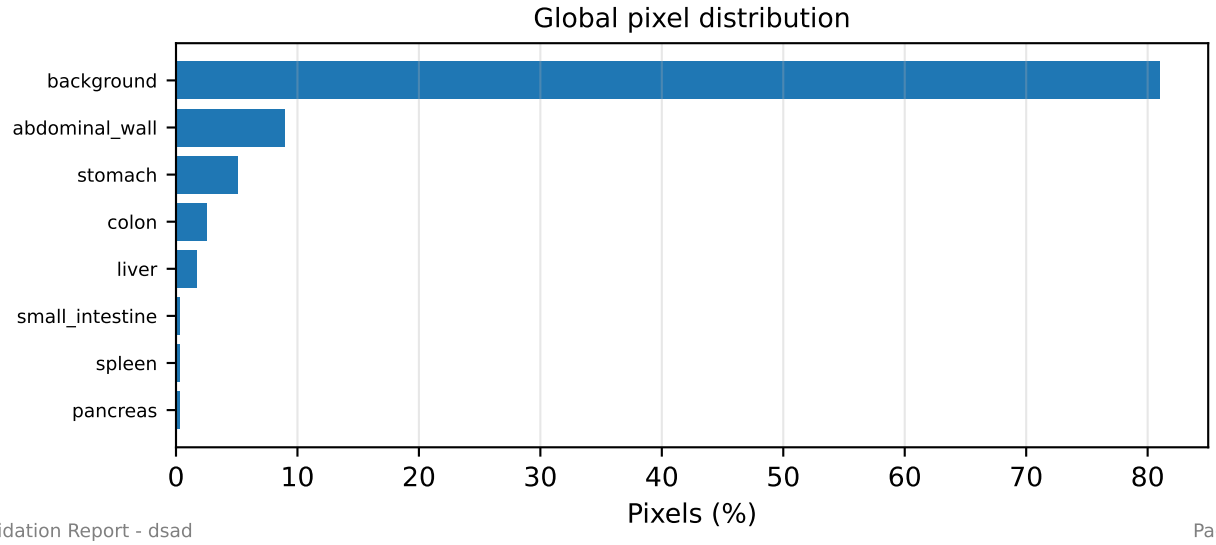
Classes

```
0  background
1  abdominal_wall
2  colon
3  liver
4  pancreas
5  small_intestine
6  spleen
7  stomach
```

3. Global Statistics

All splits together - rows highlighted = class imbalance warning (part 1)

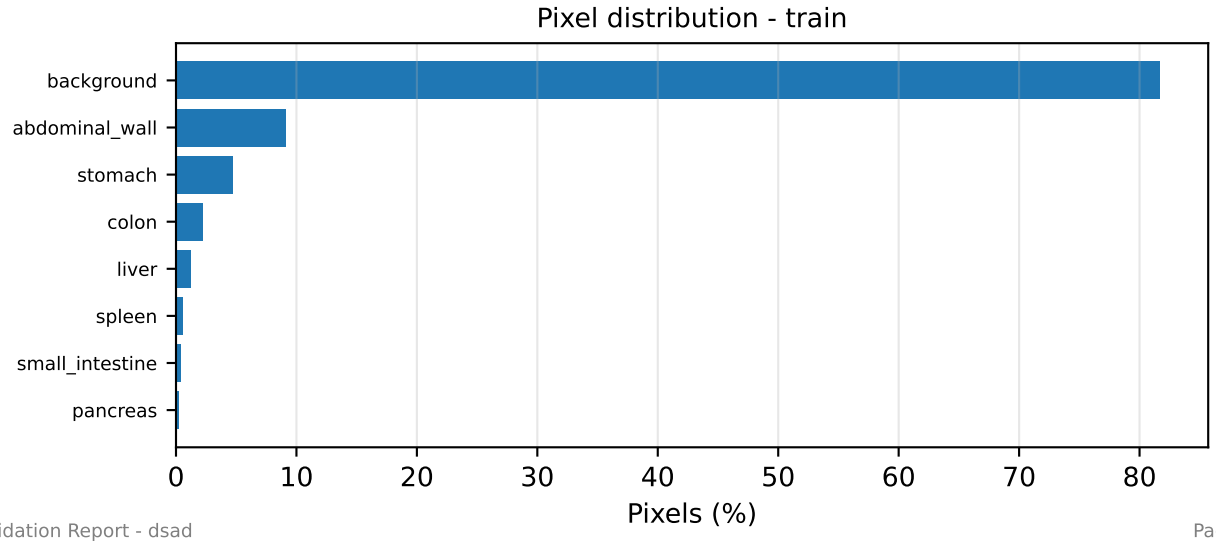
Class	Frames	Frames %	Pixels	Pixels %	Videos
background	1,430	100.00	1,517,066,422	80.9391	23/23
abdominal_wall	965	67.48	166,851,112	8.9019	23/23
colon	712	49.79	47,606,514	2.5399	22/23
liver	236	16.50	31,705,185	1.6915	10/23
pancreas	208	14.55	4,984,008	0.2659	13/23
small_intestine	78	5.45	6,056,299	0.3231	6/23
spleen	108	7.55	5,659,162	0.3019	5/23
stomach	1,430	100.00	94,400,898	5.0365	23/23



4. Split Statistics - train

863 frames, 15 videos - highlighted = class in <= 3 videos (part 1)

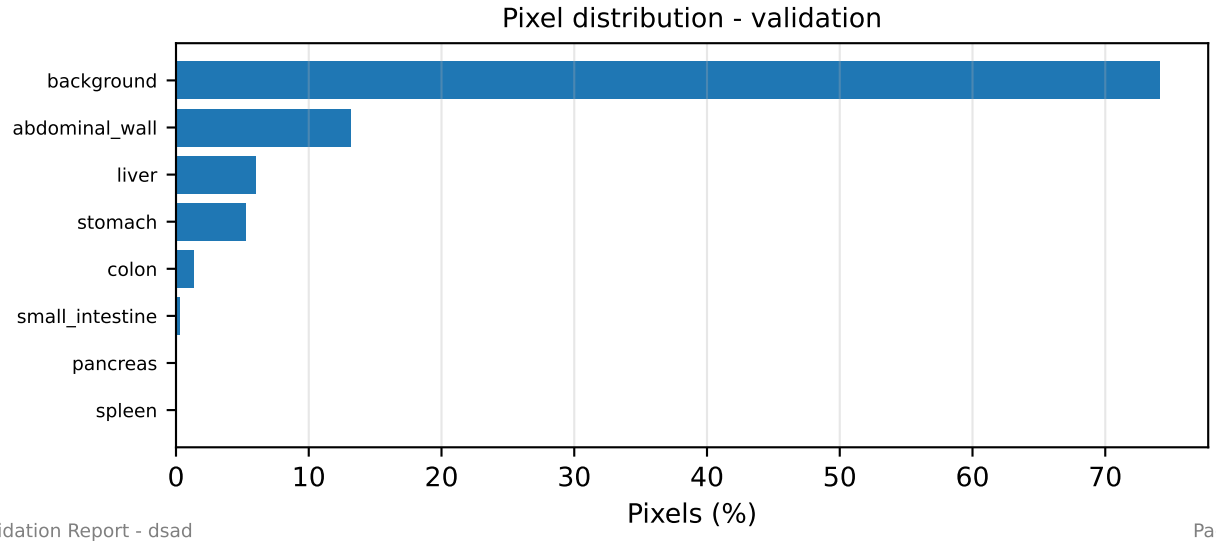
Class	Frames	Frames %	Pixels	Pixels %	Videos
background	863	100.00	923,230,982	81.6187	15/15
abdominal_wall	491	56.89	102,713,874	9.0805	15/15
colon	391	45.31	24,907,744	2.2020	14/15
liver	101	11.70	13,796,672	1.2197	5/15
pancreas	119	13.79	2,798,809	0.2474	8/15
small_intestine	57	6.60	4,609,369	0.4075	3/15
spleen	108	12.51	5,659,162	0.5003	5/15
stomach	863	100.00	53,434,748	4.7239	15/15



4. Split Statistics - validation

202 frames, 3 videos - highlighted = class in <= 3 videos (part 1)

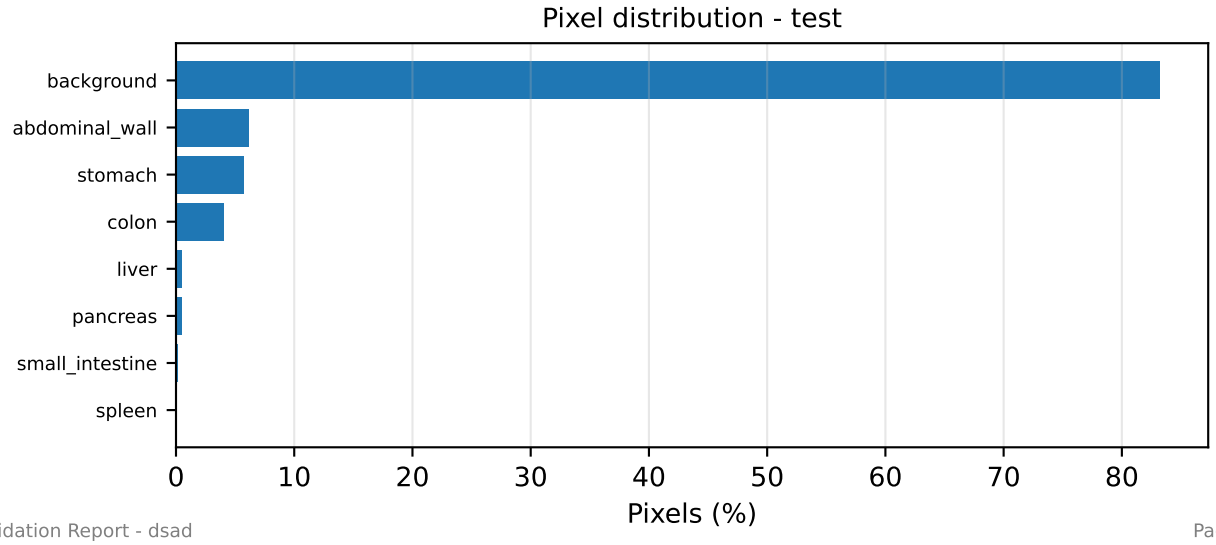
Class	Frames	Frames %	Pixels	Pixels %	Videos
background	202	100.00	196,063,774	74.0519	3/3
abdominal_wall	184	91.09	34,864,733	13.1682	3/3
colon	73	36.14	3,481,432	1.3149	3/3
liver	79	39.11	15,767,107	5.9551	2/3
pancreas	14	6.93	99,303	0.0375	1/3
small_intestine	17	8.42	714,571	0.2699	2/3
spleen	0	0.00	0	0.0000	0/3
stomach	202	100.00	13,774,520	5.2025	3/3



4. Split Statistics - test

365 frames, 5 videos - highlighted = class in <= 3 videos (part 1)

Class	Frames	Frames %	Pixels	Pixels %	Videos
background	365	100.00	397,771,666	83.1440	5/5
abdominal_wall	290	79.45	29,272,505	6.1187	5/5
colon	248	67.95	19,217,338	4.0169	5/5
liver	56	15.34	2,141,406	0.4476	3/5
pancreas	75	20.55	2,085,896	0.4360	4/5
small_intestine	4	1.10	732,359	0.1531	1/5
spleen	0	0.00	0	0.0000	0/5
stomach	365	100.00	27,191,630	5.6837	5/5



5. Video Coverage

In how many (and which) videos each class appears - global. Top share = % of the class pixels in its dominant video

Class	Videos	Videos %	Top video	Top share %	Which videos
background	23/23	100	12	6	02, 03, 04, 05, 07, ... (+18)
abdominal_wall	23/23	100	25	15	02, 03, 04, 05, 07, ... (+18)
colon	22/23	96	18	18	02, 03, 04, 05, 07, ... (+17)
liver	10/23	43	03	46	02, 03, 18, 20, 21, ... (+5)
pancreas	13/23	57	22	22	02, 03, 04, 05, 07, ... (+8)
small_intestine	6/23	26	25	35	03, 05, 17, 18, 21, ... (+1)
spleen	5/23	22	12	75	04, 12, 22, 23, 24
stomach	23/23	100	25	12	02, 03, 04, 05, 07, ... (+18)

5. Video Coverage - train

15 videos in this split (part 1)

Class	Videos	Top video	Top share %	Which videos
background	15/15	12	10	04, 05, 08, 10, 12, ... (+10)
abdominal_wall	15/15	25	24	04, 05, 08, 10, 12, ... (+10)
colon	14/15	31	19	04, 05, 08, 10, 12, ... (+9)
liver	5/15	25	60	23, 24, 25, 30, 31
pancreas	8/15	22	40	04, 05, 17, 22, 24, ... (+3)
small_intestine	3/15	25	46	05, 17, 25
spleen	5/15	12	75	04, 12, 22, 23, 24
stomach	15/15	25	22	04, 05, 08, 10, 12, ... (+10)

5. Video Coverage - validation

3 videos in this split (part 1)

Class	Videos	Top video	Top share %	Which videos
background	3/3	03	39	03, 21, 26
abdominal_wall	3/3	26	55	03, 21, 26
colon	3/3	03	34	03, 21, 26
liver	2/3	03	93	03, 21
pancreas	1/3	03	100	03
small_intestine	2/3	03	78	03, 21
spleen	0/3	-	0	-
stomach	3/3	03	59	03, 21, 26

5. Video Coverage - test

5 videos in this split (part 1)

Class	Videos	Top video	Top share %	Which videos
background	5/5	11	24	02, 07, 11, 18, 20
abdominal_wall	5/5	20	29	02, 07, 11, 18, 20
colon	5/5	18	44	02, 07, 11, 18, 20
liver	3/5	02	69	02, 18, 20
pancreas	4/5	02	34	02, 07, 11, 18
small_intestine	1/5	18	100	18
spleen	0/5	-	0	-
stomach	5/5	11	34	02, 07, 11, 18, 20

5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 1/2, videos part 1/1)

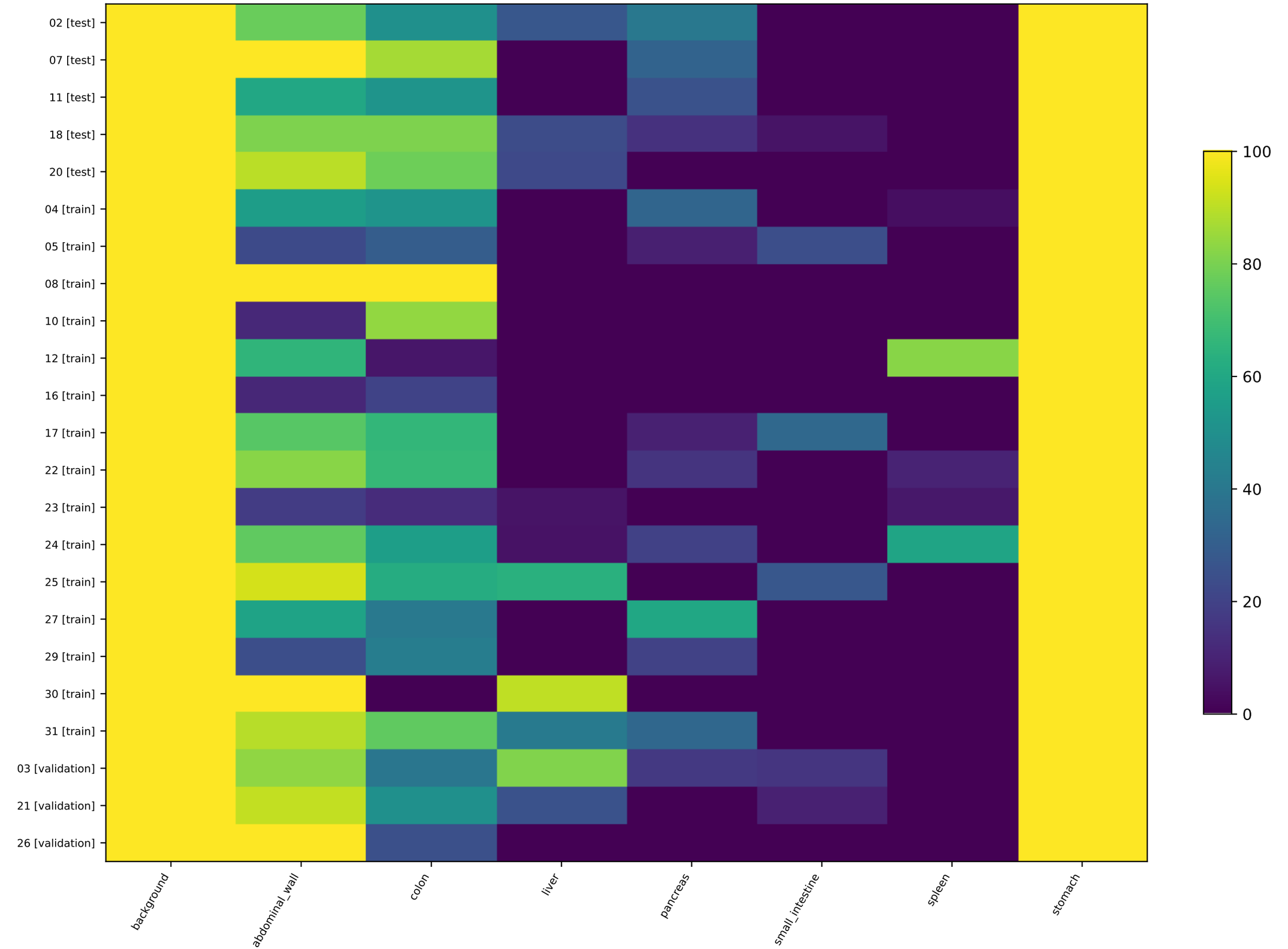
	background		abdominal_wall		colon		liver		pancreas		small_intestine	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
02 [test]	70	76.45M	54	5.61M	35	2.57M	19	1.48M	28	717.7k	0	0
07 [test]	44	51.47M	44	3.20M	38	1.49M	0	0	14	91.6k	0	0
11 [test]	87	95.53M	52	6.06M	45	2.61M	0	0	22	645.1k	0	0
18 [test]	78	82.57M	63	5.88M	63	8.48M	18	191.9k	11	631.6k	4	732.4k
20 [test]	86	91.76M	77	8.51M	67	4.07M	19	470.8k	0	0	0	0
04 [train]	52	64.16M	29	2.75M	27	194.2k	0	0	17	244.7k	0	0
05 [train]	58	70.14M	13	1.55M	17	1.64M	0	0	5	125.4k	14	599.4k
08 [train]	18	17.55M	18	3.58M	18	2.19M	0	0	0	0	0	0
10 [train]	43	54.61M	5	158.0k	36	975.9k	0	0	0	0	0	0
12 [train]	84	96.19M	55	5.90M	5	35.7k	0	0	0	0	0	0
16 [train]	45	55.03M	5	209.7k	9	415.0k	0	0	0	0	0	0
17 [train]	77	85.27M	57	6.26M	51	2.43M	0	0	7	117.9k	26	1.88M
22 [train]	79	77.80M	65	12.04M	53	4.22M	0	0	12	1.12M	0	0
23 [train]	78	93.93M	14	3.63M	10	1.01M	4	192.2k	0	0	0	0
24 [train]	41	42.72M	31	5.70M	23	2.65M	2	47.7k	8	167.9k	0	0
25 [train]	63	35.63M	59	24.27M	39	507.7k	40	8.27M	0	0	17	2.13M
27 [train]	57	64.29M	33	6.94M	23	1.52M	0	0	34	277.7k	0	0
29 [train]	71	82.01M	17	938.2k	30	2.30M	0	0	14	419.6k	0	0
30 [train]	31	14.94M	31	19.79M	0	0	28	3.97M	0	0	0	0
31 [train]	66	68.97M	59	8.97M	50	4.82M	27	1.32M	22	329.8k	0	0
03 [validation]	84	75.84M	70	9.64M	33	1.18M	68	14.69M	14	99.3k	13	554.7k
21 [validation]	44	46.79M	40	6.07M	22	1.15M	11	1.07M	0	0	4	159.9k
26 [validation]	74	73.43M	74	19.15M	18	1.14M	0	0	0	0	0	0
TOTAL	**1,430**	**1517.07M**	**965**	**166.85M**	**712**	**47.61M**	**236**	**31.71M**	**208**	**4.98M**	**78**	**6.06M**

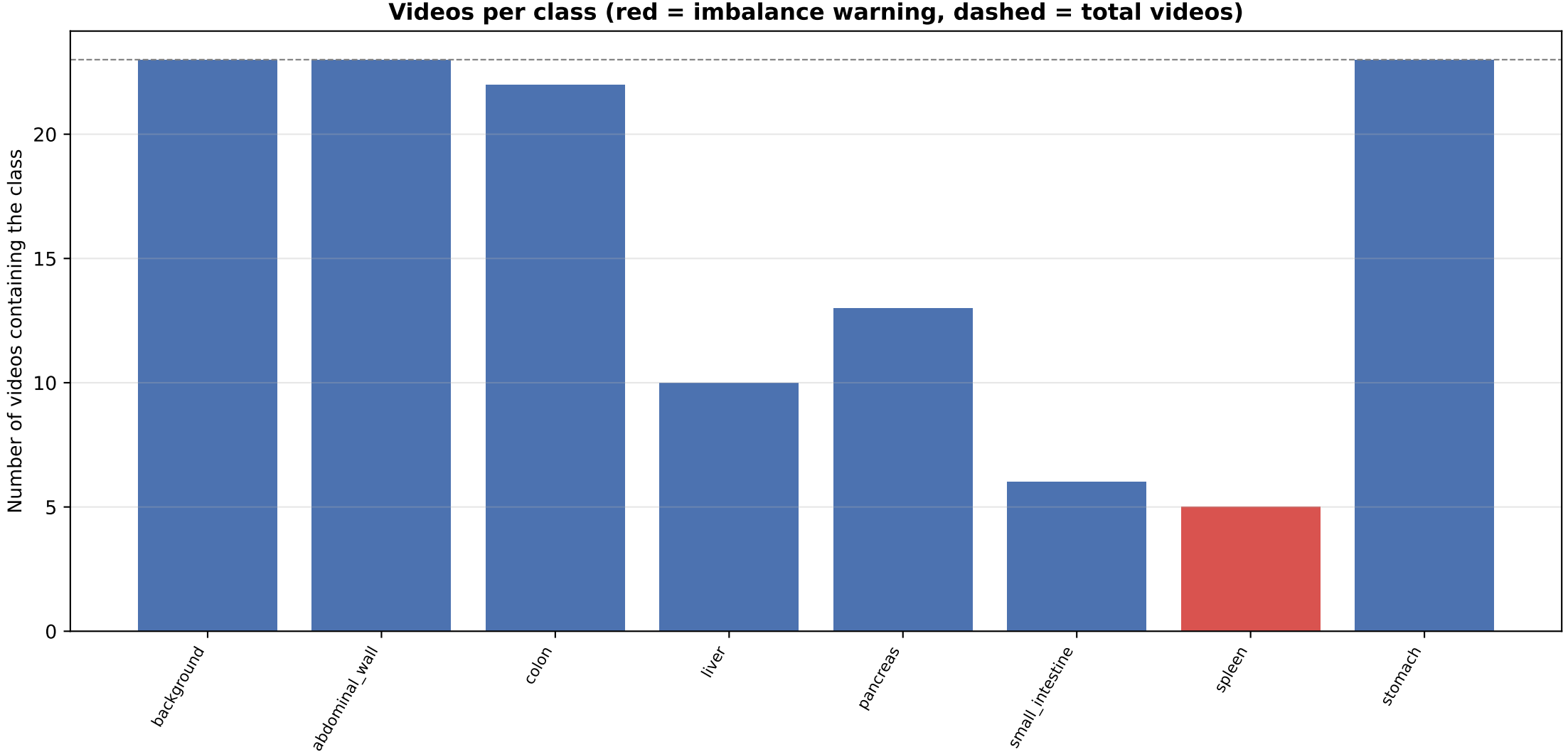
5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 2/2, videos part 1/1)

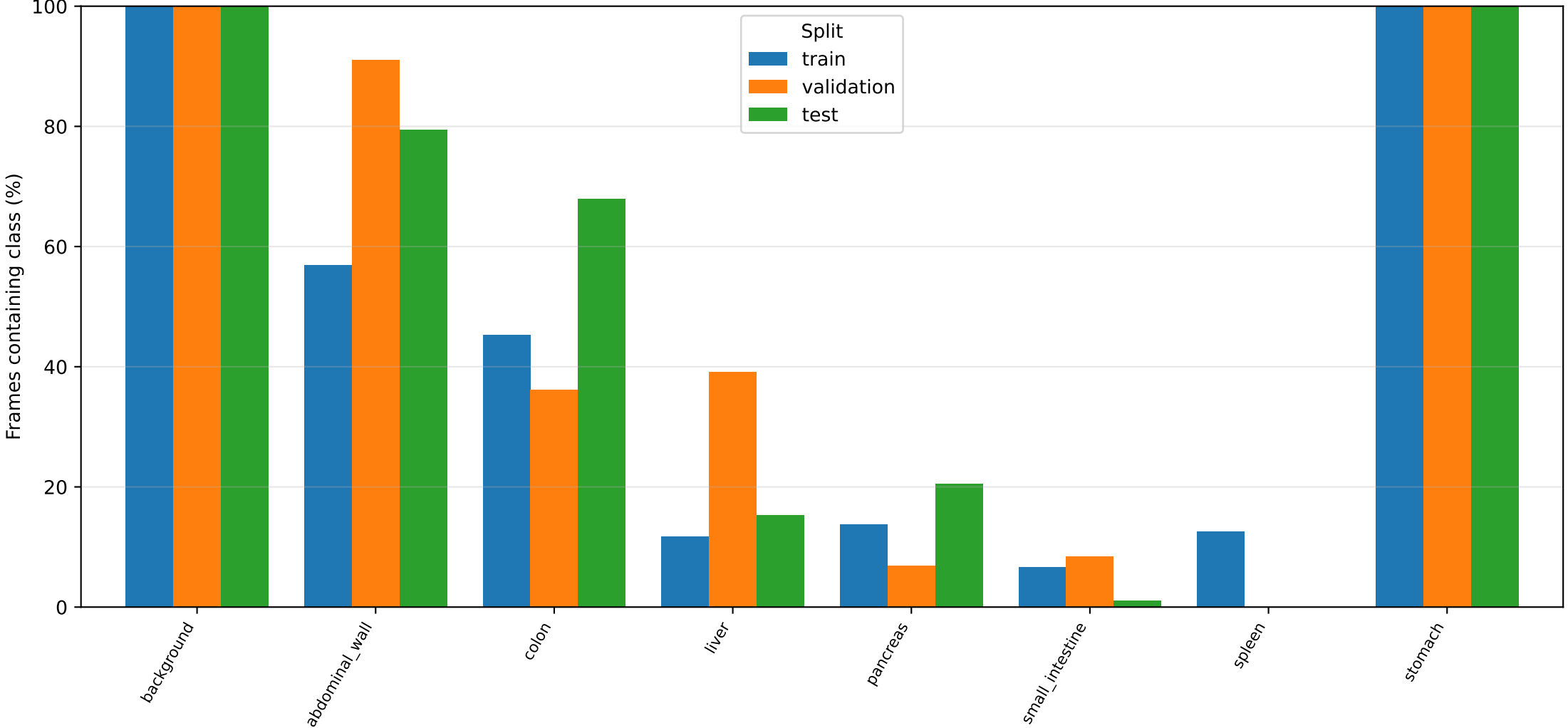
	spleen		stomach	
Video [split]	**frames**	**pixels**	**frames**	**pixels**
02 [test]	0	0	70	4.93M
07 [test]	0	0	44	1.41M
11 [test]	0	0	87	9.19M
18 [test]	0	0	78	3.75M
20 [test]	0	0	86	7.91M
04 [train]	2	17.7k	52	790.0k
05 [train]	0	0	58	1.96M
08 [train]	0	0	18	269.6k
10 [train]	0	0	43	612.7k
12 [train]	69	4.27M	84	3.71M
16 [train]	0	0	45	3.33M
17 [train]	0	0	77	4.96M
22 [train]	8	51.0k	79	8.32M
23 [train]	5	27.9k	78	3.44M
24 [train]	24	1.29M	41	1.17M
25 [train]	0	0	63	11.76M
27 [train]	0	0	57	1.69M
29 [train]	0	0	71	7.40M
30 [train]	0	0	31	1.94M
31 [train]	0	0	66	2.10M
03 [validation]	0	0	84	8.08M
21 [validation]	0	0	44	2.42M
26 [validation]	0	0	74	3.27M
TOTAL	**108**	**5.66M**	**1,430**	**94.40M**

Class presence per video (% of the video's frames containing the class)

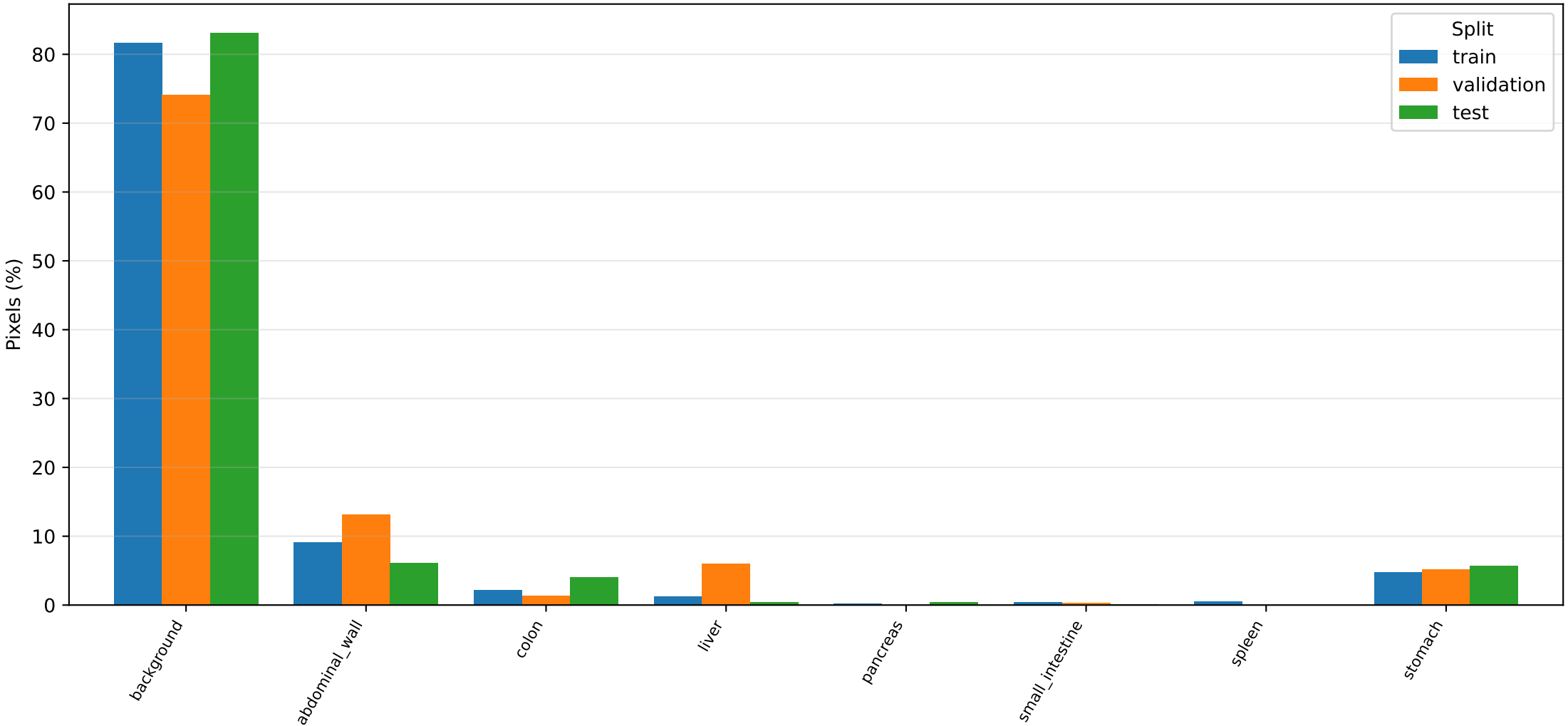



Videos per class (red = imbalance warning, dashed = total videos)
Number of videos containing the class
0
5
10
15
20
background
abdominal_wall
colon
liver
pancreas
small_intestine
spleen
stomach

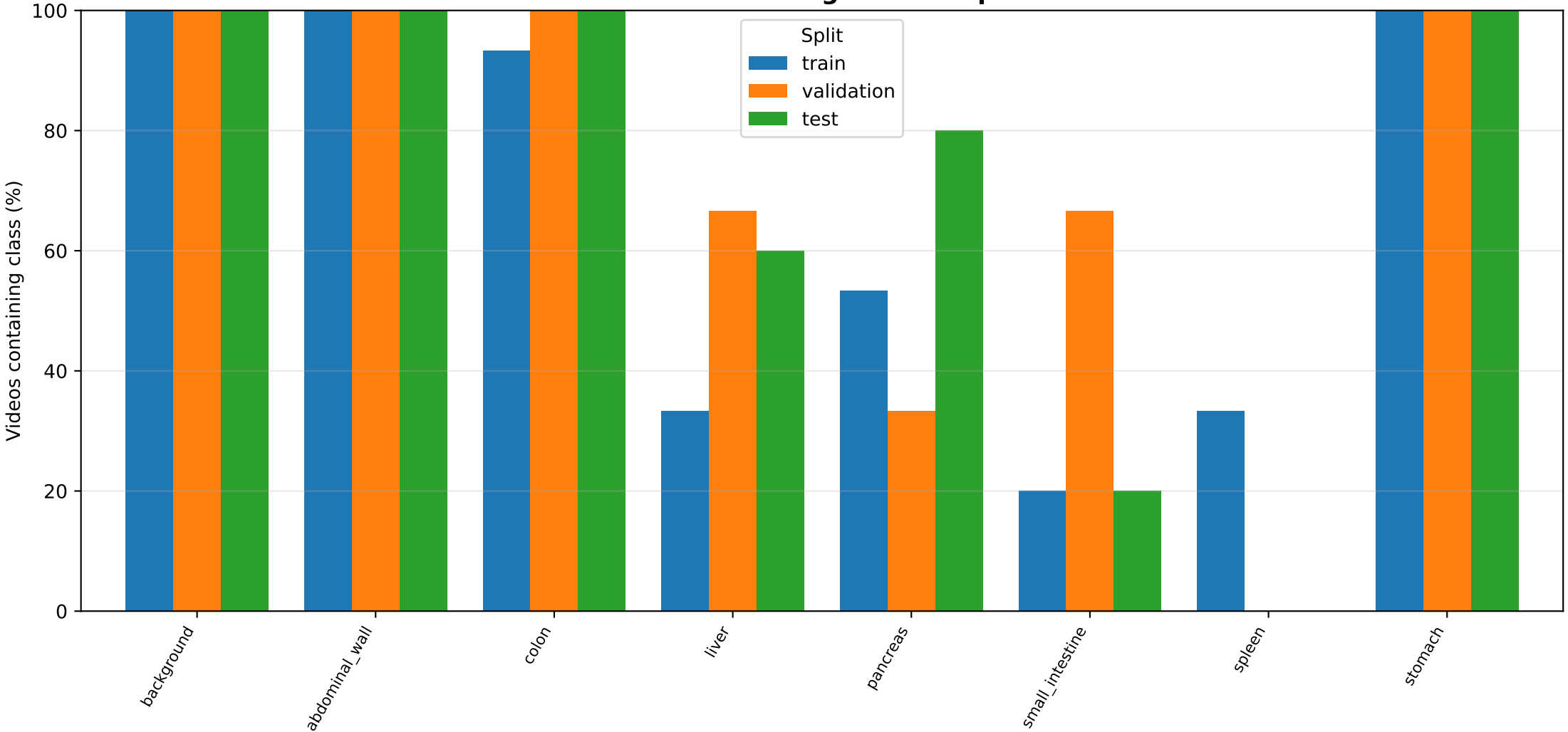
6. Class presence across splits



6. Pixel distribution across splits

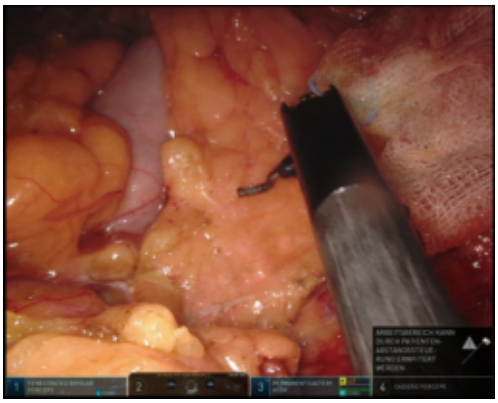


6. Video coverage across splits



7. Qualitative Analysis - train

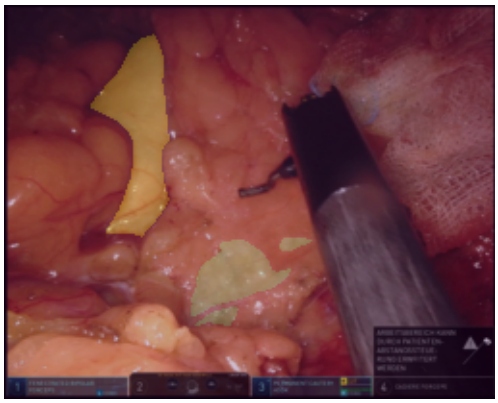
Image - #562 (24)



Ground truth

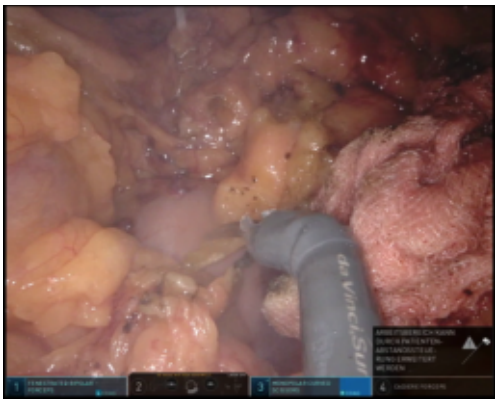


Overlay



Classes: background, pancreas, stomach

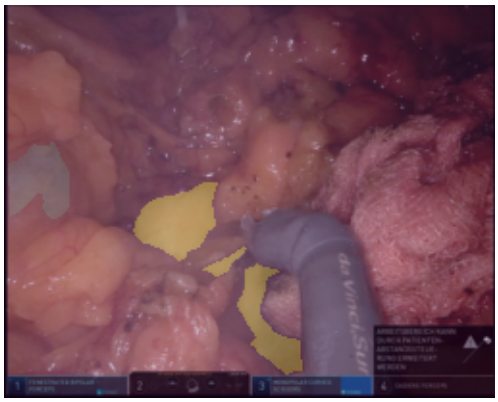
Image - #664 (27)



Ground truth

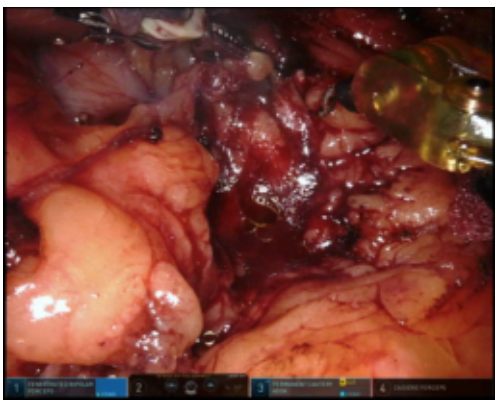


Overlay



Classes: background, colon, stomach

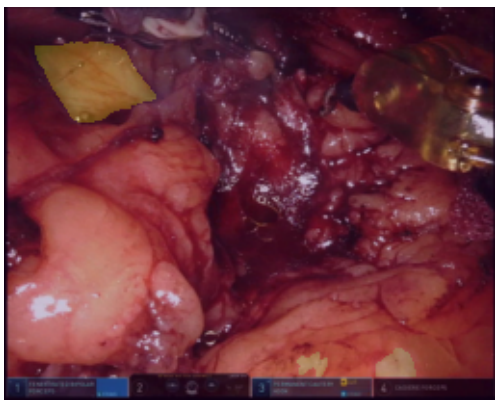
Image - #373 (17)



Ground truth



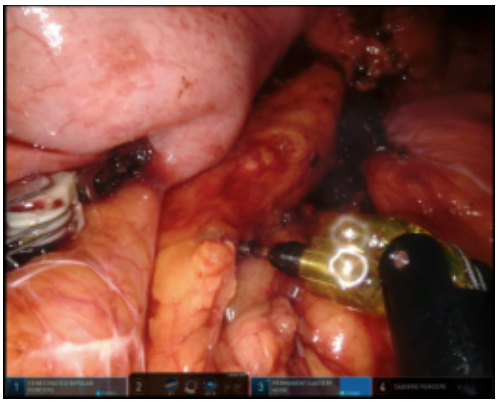
Overlay



Classes: background, pancreas, stomach

7. Qualitative Analysis - train

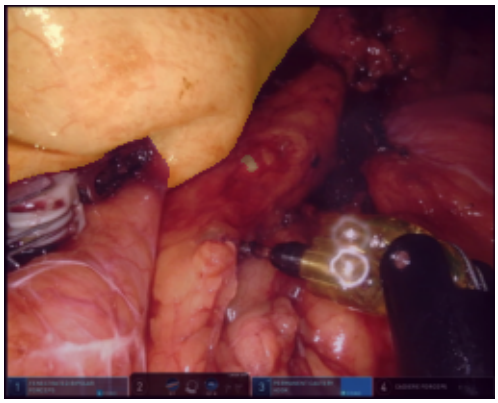
Image - #740 (29)



Ground truth

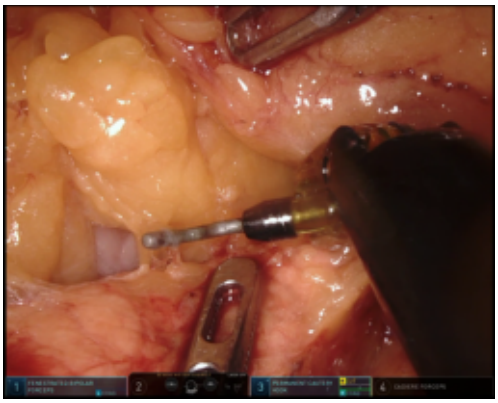


Overlay



Classes: background, pancreas, stomach

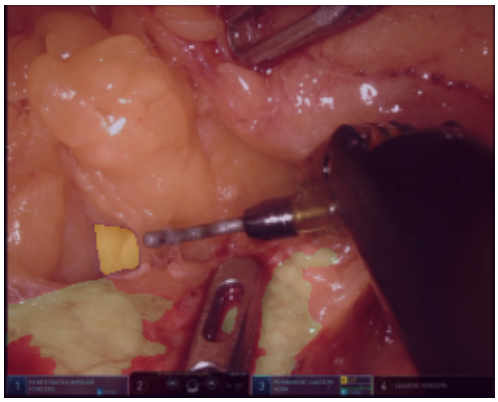
Image - #377 (22)



Ground truth

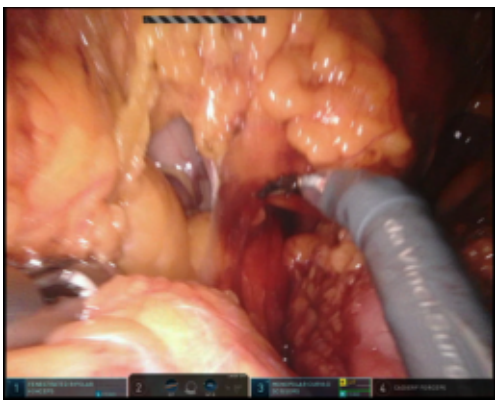


Overlay



Classes: background, pancreas, stomach

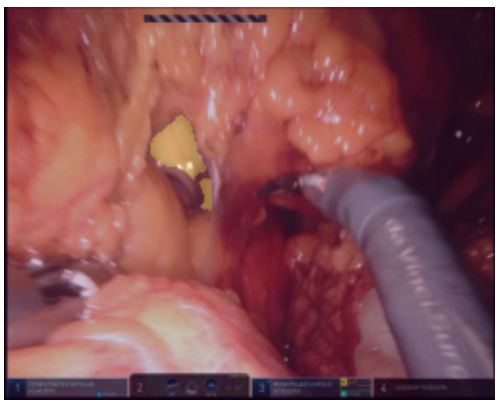
Image - #76 (05)



Ground truth



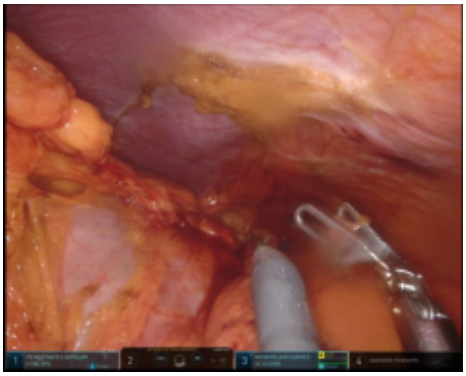
Overlay



Classes: background, colon, stomach

7. Qualitative Analysis - validation

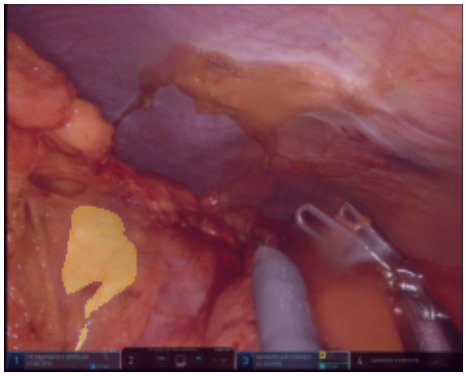
Image - #993 (26)



Ground truth

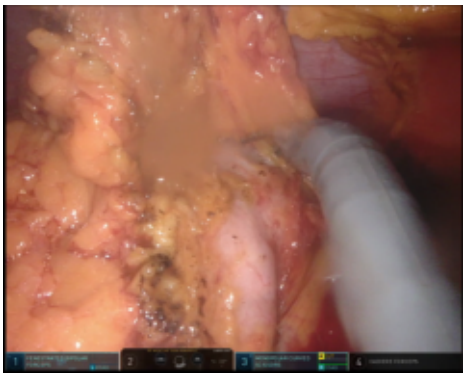


Overlay



Classes: background, abdominal_wall, stomach

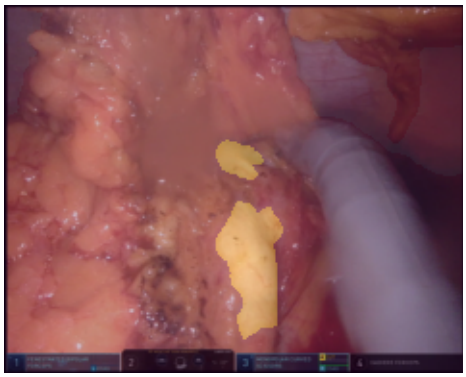
Image - #1016 (26)



Ground truth

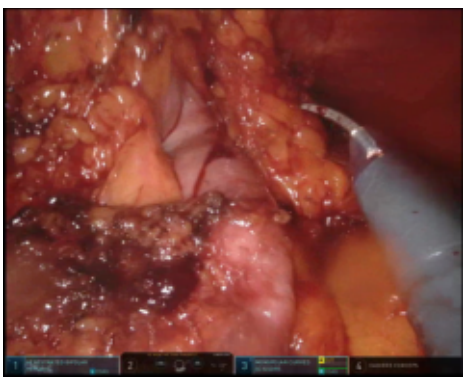


Overlay



Classes: background, abdominal_wall, stomach

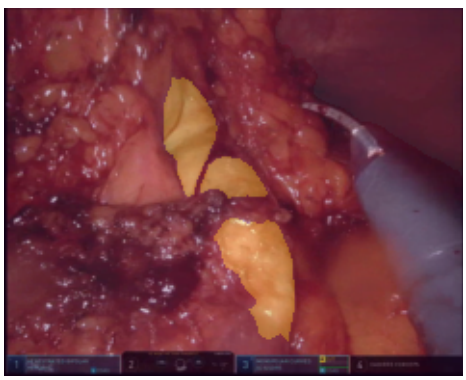
Image - #1063 (26)



Ground truth



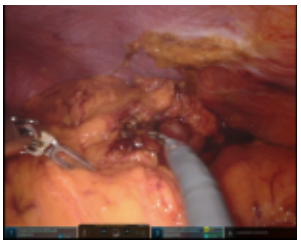
Overlay



Classes: background, abdominal_wall, stomach

7. Qualitative Analysis - validation

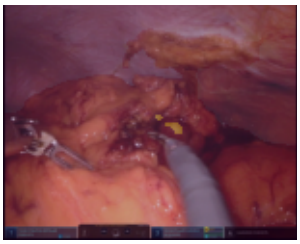
Image - #1036 (26)



Ground truth

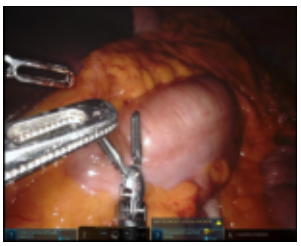


Overlay



Classes: background, abdominal_wall, stomach

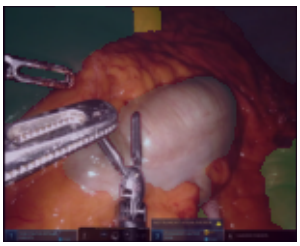
Image - #950 (21)



Ground truth

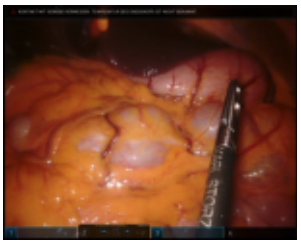


Overlay



Classes: background, abdominal_wall, colon, liver, small_intestine, stomach

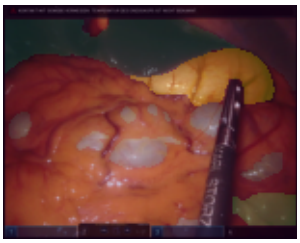
Image - #880 (03)



Ground truth



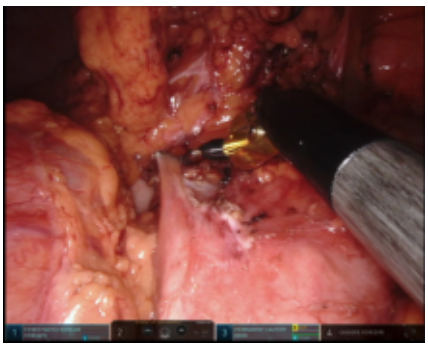
Overlay



Classes: background, abdominal_wall, colon, liver, small_intestine, stomach

7. Qualitative Analysis - test

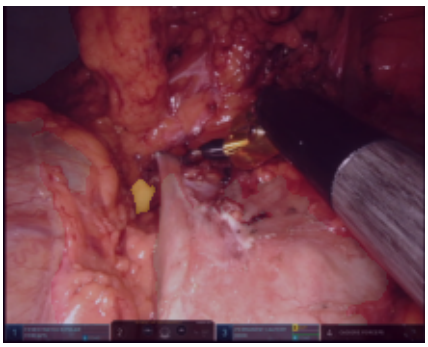
Image - #1301 (18)



Ground truth

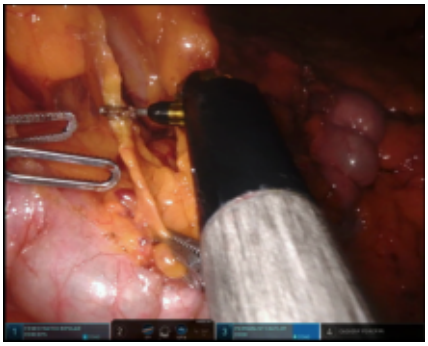


Overlay



Classes: background, abdominal_wall, colon, stomach

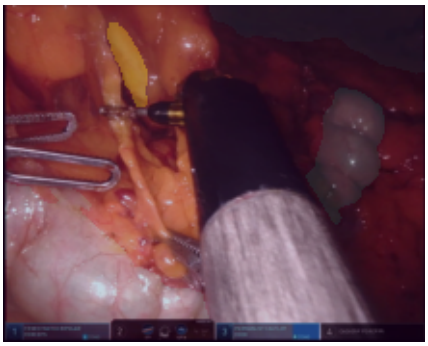
Image - #1344 (20)



Ground truth

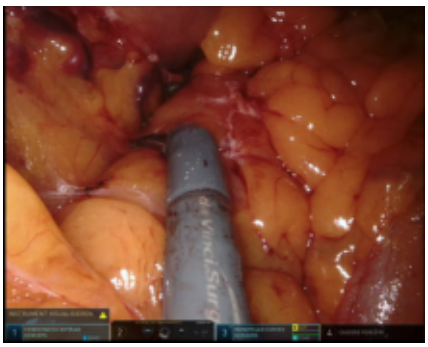


Overlay



Classes: background, abdominal_wall, colon, stomach

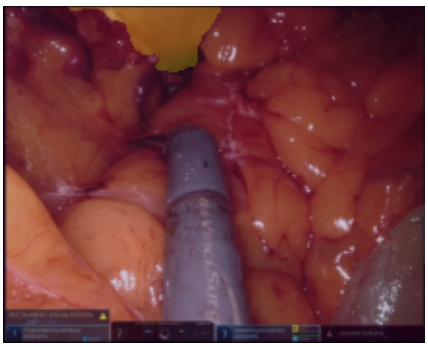
Image - #1222 (11)



Ground truth



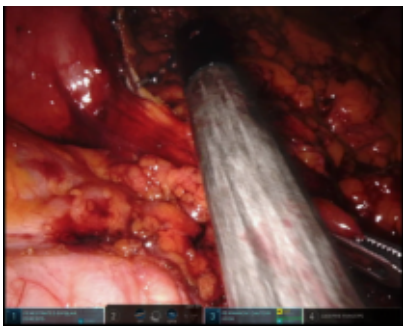
Overlay



Classes: background, colon, stomach

7. Qualitative Analysis - test

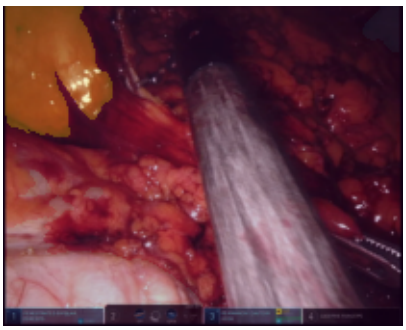
Image - #1378 (20)



Ground truth

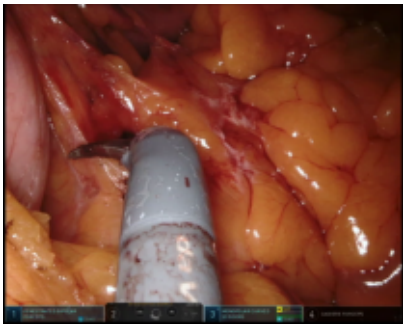


Overlay



Classes: background, abdominal_wall, colon, stomach

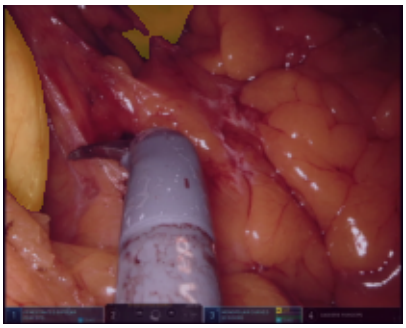
Image - #1224 (11)



Ground truth

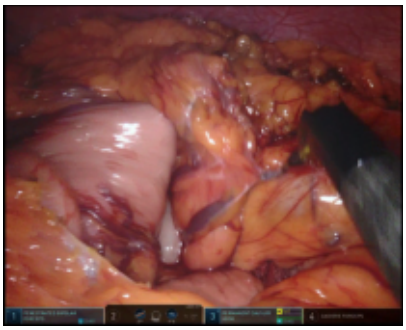


Overlay



Classes: background, stomach

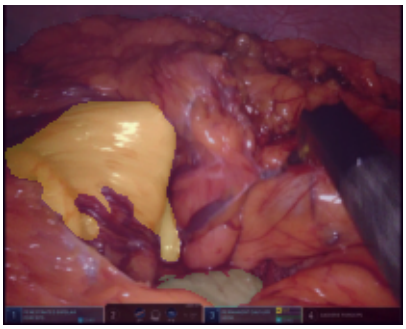
Image - #1097 (02)



Ground truth



Overlay



Classes: background, abdominal_wall, pancreas, stomach

8. Conclusions

Dataset: dsad

Frames, pixels and videos measure three different things:
- Frames: how often the class occurs across images.
- Pixels: how much image area the class occupies.
- Videos: how many independent patients/procedures show the class.

A class present in many frames but in only a few videos is a weak basis for generalisation: frames of the same video are highly correlated, so the effective sample size is closer to the number of videos than to the number of frames.

Splits are made at video level, so a rare class can end up entirely in train or entirely in validation/test. Check 'Classes missing from a split' in the executive summary before interpreting per-class metrics.

Classes flagged in red (few videos, or concentrated in one video) deserve special care: per-class metrics on them are high-variance, and class weighting or oversampling should be evaluated.